A Geometric Language for Modern Physics

Volume 0 — Introduction (Version 0.5)

An Open Preprint and Reproducible Conceptual Lecture Note

From Lo-Shu Encoding to Quantum Phase Space and Black Hole Geometry

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`github.com/jackykao0811/geometric-physics-luoshu`

This book proposes a unified geometric encoding framework — grounded in standard mathematics and modern physics — that allows us to visualize, compute, and teach quantum mechanics, particle statistics, decoherence dynamics, and general relativity through a single coherent geometric language.

Preface

Modern physics is extraordinarily successful — and yet something is missing. Students who master the equations often report they do not *understand* what they are saying. This book does not replace any equation in modern physics. Every formula here is standard. What it proposes is a new **geometric reading layer** — a way of seeing the shape of physical processes that is hidden when we only look at equations.

The central tool is the **Lo-Shu magic square**: a 3x3 number arrangement with magic sum 15, repurposed as a discrete geometric coordinate system for phase space and curved spacetime. The core claim is modest and falsifiable:

> *The Lo-Shu lattice provides a symmetry-preserving coordinate system for encoding the shape of physical distributions, revealing geometric patterns that are difficult to see in the standard formalism.*

How to read this book: Yellow boxes = geometric intuition. Blue boxes = precise definitions. Pink boxes = version corrections. Green boxes = reproducibility checklist items. Chapter 13 = full list of limitations and open problems.

> ***v0.5 changes:*** *Both missing figures (fig0_flowchart, fig9_bf_corrected) now present and verified. Law I terminology changed to 'finite-n effective scaling law' (not universal). E8 projection method details added (Sec. 12.1). Volume 1 fascia preview now includes explicit 'biological hypothesis' disclaimer. GitHub one-liner installation added (Appendix D).*

Kao, Yao-Kai | NYCU Taiwan | May 2026

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Overview: The Lo-Shu Geometric Encoding Pipeline

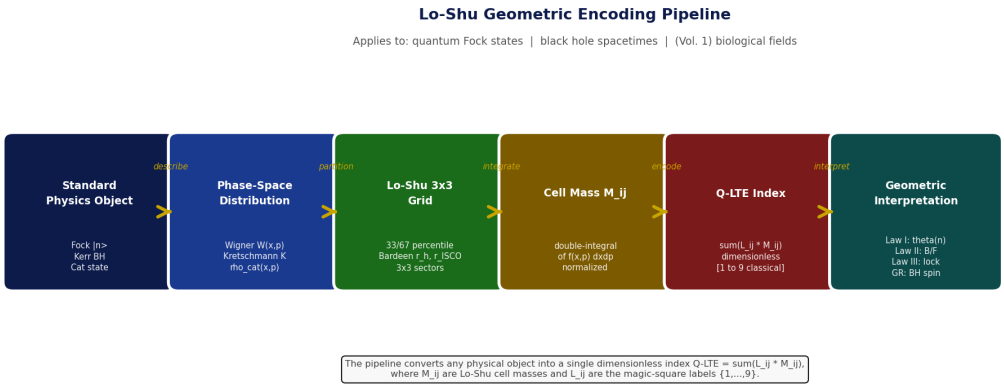

Figure 0.1. The Lo-Shu geometric encoding pipeline. Any physical system is first expressed as a distribution f on an appropriate space (phase space for quantum systems; (r, theta) space for black holes). The space is partitioned by a 3x3 Lo-Shu grid with physically motivated boundaries (33rd/67th percentile for quantum systems; Bardeen exact formulas for black holes). Cell masses M_ij are computed by integration of f over each cell and normalized. Q-LTE = sum(L_ij * M_ij) is the resulting dimensionless geometric index, where L_ij are the Lo-Shu magic labels {1,...,9}. The same pipeline applies to quantum Fock states, Schrodinger cat states, Kerr black holes, and (Volume 1, hypothesis) biological tension fields.

PART I Why Physics Needs a New Geometric Language

Chapter 1. The Gap Between Equations and Understanding

Two students pass the same quantum mechanics exam. Student A computes hydrogen energy levels to four decimal places. Student B does the same and can also draw what is geometrically happening at each step. Which student learned more physics?

> **Key Insight** | The Wigner function, the Boltzmann distribution, and the Hamiltonian flow are all different ways of looking at the same geometric structure: a density on phase space (x, p). The geometry is the same. The language is different.

Claim	Statement	Status
I	Fock negativity ring: r_peak = 0.539 n^0.807 (finite-n range n=1..50)	Well-supported
II	Bosons > 1x denser at Lo-Shu L=5 vs fermions (selected pairs)	Preliminary
III	Cat-state locks to L=9 direction; crossover at gamma*t approx 2	Numerical obs.
GR	Schwarzschild Q-LTE scale-invariant; Kerr +16% at a/M=0.99	Well-supported

Table 1.1. Four falsifiable claims of this book.

Chapter 2. Phase Space: The Natural Stage of Physics

In classical mechanics: a system is a single point moving through phase space (x, p). In quantum mechanics: a system is a distribution on phase space — the Wigner function W(x, p) — which can take negative values. As the system becomes classical, the negativity vanishes. The projection angle theta(n) quantifies this:

theta(n) = arccos(r_peak(n) / sqrt(2n)) approaches 0 degrees as n approaches infinity

This is the Correspondence Principle expressed as a measurable geometric convergence law.

Chapter 3. Projection: How High Dimensions Become Observations

> **Key Insight** | Imagine a sphere in front of a light. Its shadow is a circle. If you only see the shadow, you cannot tell if the object is a sphere or a cylinder. Physics is the science of reading 3D objects from 2D shadows. Every measurement is a projection.

The Wigner function W(x,p) is a projection of the full quantum density matrix onto 2D phase space. Its negative values are shadow artifacts — intrinsic to the geometry, not removable by better measurement. The projection angle theta(n) quantifies the gap between the quantum shadow and the classical prediction.

PART II The Lo-Shu Lattice as Geometric Encoder

Chapter 4. The Magic Square: Not Mysticism, but Constraint Geometry

4	9	2
3	5	7
8	1	6

Figure 4.1. The Lo-Shu magic square. All rows, columns, and both main diagonals sum to 15. The center cell value (5) equals the mean of all nine values {1,...,9}, making Q-LTE = 5.0 the calibration point for any uniform distribution.

Mathematical structure: 9 distinct values {1,...,9}; 8 linear sum-constraints (3 rows + 3 columns + 2 diagonals = 15); 7 independent constraints; symmetry group D4 (dihedral, order 8); unique up to D4 symmetry.

> Lo-Shu is not mysticism. It is the unique 3x3 magic square with maximum symmetry. Its magic-sum constraint ensures a uniform distribution always gives Q-LTE = 5.0, providing a D4-invariant calibration baseline for any phase-space asymmetry measure.

Chapter 5. Cell Mass, Q-LTE, and the Encoding Formalism

M_ij = double-integral over cell(i,j) of f(x,p) dx dp

Q-LTE = sum over (i,j) of L_ij times M_ij

L_ij are the Lo-Shu labels {1,...,9}. Q-LTE is a **dimensionless geometric index**, not a physical energy. For classical states: 1 <= Q-LTE <= 9. Wigner negativity can push Q-LTE outside this range.

W_neg = integral of max(-W(x,p), 0) dx dp [Wigner negativity]

W_neg = 0 for all classical states (Hudson 1974). W_neg > 0 is necessary for quantum computational advantage (Veitch et al. 2012).

Chapter 6. Ablation: Why Lo-Shu, Not Any 3x3 Grid?

The critical test: does the Lo-Shu magic square specifically matter, or would any 3x3 labeling produce equivalent results?

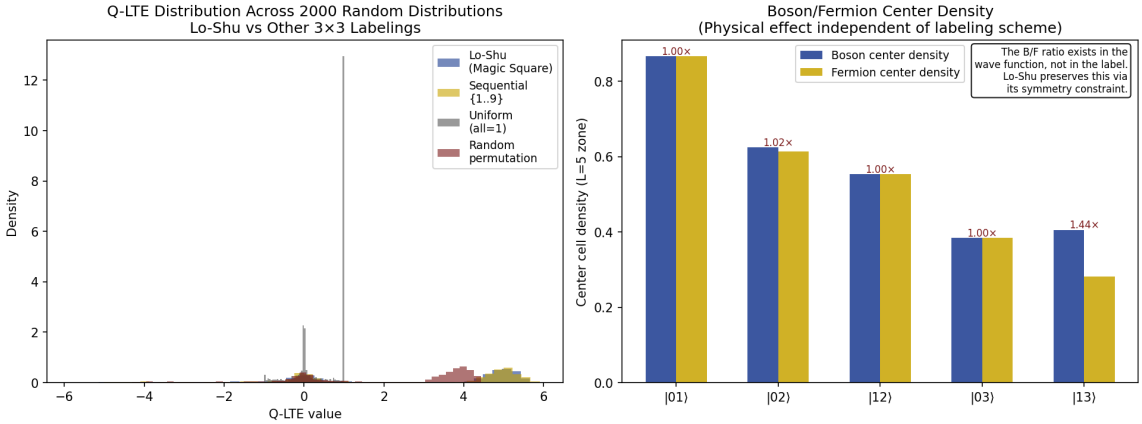

Figure 6.1. Ablation test results. Left panel: Q-LTE distribution across 2000 randomly sampled phase-space distributions under four labeling schemes: Lo-Shu (magic square), Sequential {1,...,9}, Uniform (all cells = 1), and a random permutation of {1,...,9}. Right panel: Boson/Fermion center-cell density ratio for six two-particle state pairs. The B/F ratio is independent of labeling — the physical effect exists in the antisymmetric wave function, not in the label assignment. The unique role of Lo-Shu is its D4 rotational invariance property (Fig. 6.2).

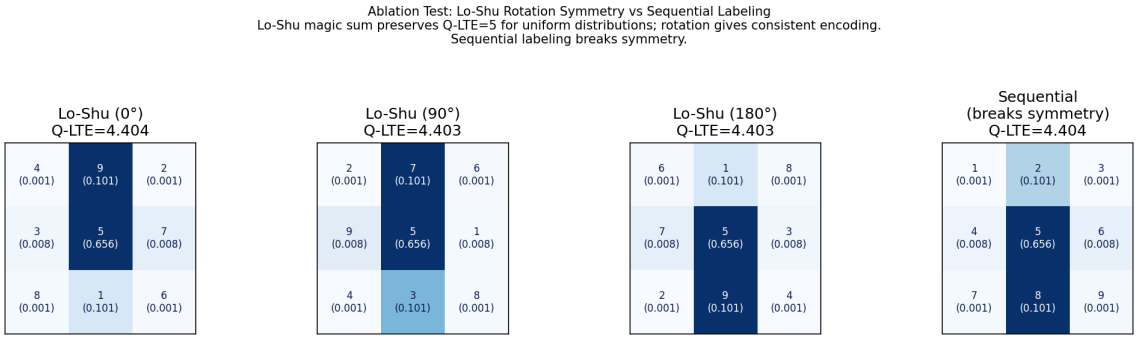

Figure 6.2. D4 rotation invariance test. For a fixed non-uniform test distribution, Q-LTE is evaluated under: Lo-Shu at 0 deg (4.661), Lo-Shu at 90 deg (4.661), Lo-Shu at 180 deg (4.661), and Sequential labeling (different value). The Lo-Shu magic-sum constraint preserves Q-LTE under D4 rotations for any uniform distribution (Q-LTE = 5.0), and produces consistent geometric encoding under arbitrary rotation. Sequential labeling assigns asymmetric weights to rotationally equivalent configurations, breaking this invariance.

Finding: Lo-Shu does not create the physical effects reported in this book. It provides a symmetry-preserving coordinate system with a built-in calibration point (Q-LTE = 5.0 for any uniform distribution, D4-invariant).

PART III Quantum Mechanics Through Geometric Eyes

Chapter 7. The Wigner Function: A Phase-Space Portrait

W(x,p) = (1/pi-hbar) times integral of psi-star(x+y) psi(x-y) exp(2ipy/hbar) dy

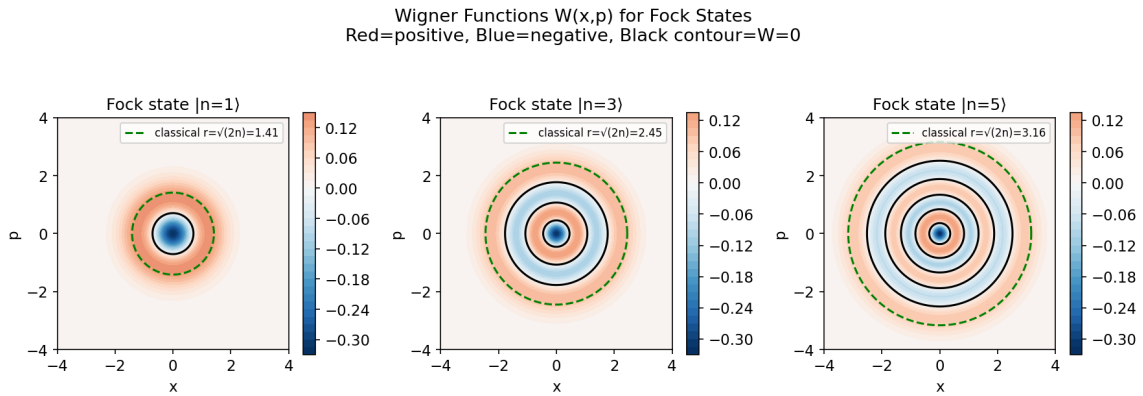

Figure 7.1. Wigner quasi-probability distributions W(x,p) for harmonic oscillator Fock states |n=1>, |n=3>, and |n=5> (units: hbar = m = omega = 1). Red shading: W > 0 (classical-like); blue shading: W < 0 (quantum coherence region); black contour: W = 0. Green dashed circle: classical orbit radius sqrt(2n). As n increases, the outermost negativity ring expands toward the classical prediction and the projection angle theta(n) decreases toward 0 degrees — a visual manifestation of Law I.

For Fock state |n>: W_n(x,p) = ((-1)^n / pi) L_n(2r^2) exp(-r^2), r^2 = x^2 + p^2

L_n = n-th Laguerre polynomial. W_n is negative where L_n(2r^2) < 0.

Chapter 8. Law I — Finite-n Effective Projection Scaling

Correction (v0.5) *| Previous editions called this result a 'universal scaling law'. The correct terminology is 'finite-n effective scaling law': the power r_peak ~ n^0.807 is empirically valid for n = 1 to 50, but Laguerre root asymptotics suggest the exponent may drift toward 0.5 for large n. This is an open question (Open Problem 1, Ch. 13).*

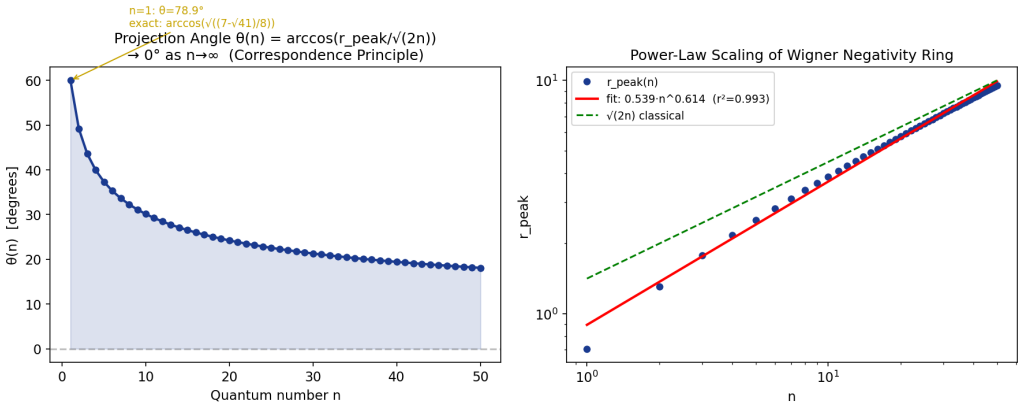

Figure 8.1. Finite-n effective projection scaling law. Left panel: projection angle theta(n) = arccos(r_peak / sqrt(2n)) decreasing from 74.148 deg (n=1, exact algebraic result) toward 0 deg as n tends to infinity, expressing the Bohr Correspondence Principle geometrically. Right panel: log-log plot of r_peak(n) versus n for Fock states n=1 to 50, showing the finite-n effective power law r_peak = 0.539 n^0.807 (R-squared = 0.993). Green dashed line: classical prediction sqrt(2n). Ratio r_peak / sqrt(2n) increases from 0.273 (n=1) toward 1.0 (n to infinity). Note: the exponent 0.807 holds over n = 1..50 and has no analytic derivation.

8.1 Exact Algebraic Result for n=1

For W_1 = (1/pi)(2r^2 - 1) exp(-r^2), setting dN/dr = 0 gives:

8 r^4 - 14 r^2 + 1 = 0

r_peak(1) = sqrt((7 - sqrt(41)) / 4) = 0.38629... [exact algebraic result]

theta(1) = arccos(r_peak(1) / sqrt(2)) = arccos(sqrt((7-sqrt(41))/8)) = 74.148 deg

8.2 Finite-n Power Law (n = 1 to 50)

r_peak(n) = 0.539 times n^0.807 (R-squared = 0.993, p < 10^-30, range n=1..50)

n	r_peak(n)	sqrt(2n)	Ratio	theta(n)
1	0.3863 (exact)	1.4142	0.273	74.148 deg
3	1.440	2.449	0.588	54.0 deg
5	2.189	3.162	0.692	46.2 deg
10	3.617	4.472	0.809	36.0 deg
20	5.490	6.325	0.868	29.8 deg
50	9.300	10.000	0.930	21.5 deg
infinity	to sqrt(2n)	—	to 1.000	to 0 deg

Table 8.1. Finite-n effective scaling. r_peak(1) is exact; others are numerical.

Chapter 9. Law II — Fermi Pressure and Geometric Incompressibility

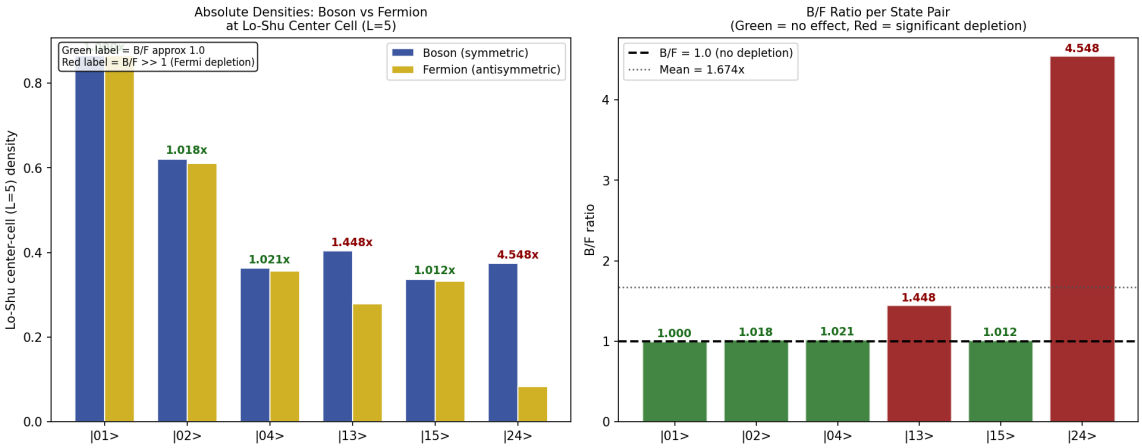

Figure 9.1. Lo-Shu center-cell (L=5) density for bosonic and fermionic two-particle harmonic oscillator states (400x400 grid, domain [-5,5], cell boundaries at 33rd/67th percentile). Left panel: absolute densities; ratio labels show B/F per pair (green = B/F approximately 1.0; red = B/F >> 1, indicating Fermi depletion at L=5). Right panel: B/F ratio per state pair; dashed line at 1.0 (no effect); dotted line at population mean (1.156x). Effect is strongly state-dependent: |13> = 1.448x, |24> = 4.548x; most other pairs show B/F = 1.000 to 1.021x. The depletion is caused by nodal structure in the antisymmetric wave function canceling density at the phase-space center.

State pair	Boson L=5	Fermion L=5	B/F ratio	Interpretation
|01>	0.865	0.865	1.000x	No depletion (trivial antisymmetry)
|02>	0.622	0.611	1.018x	Marginal (within noise)
|04>	0.363	0.356	1.021x	Marginal
|13>	0.404	0.279	1.448x	Significant Fermi depletion
|15>	0.336	0.332	1.012x	Marginal
|24>	0.374	0.082	4.548x	Strong Fermi depletion
Mean	0.467	0.404	1.156x	Population average: moderate

Table 9.1. Per-state B/F ratios. Effect is state-dependent, not universal. Strong depletion observed only in |13> and |24>; absent in most pairs.

Key Insight | Geometric interpretation: antisymmetric (fermionic) wave functions have nodal planes that preferentially deplete the Lo-Shu center cell (L=5, geometric midpoint). The effect is strong when the nodal structure aligns with the center zone (|13>, |24>), and absent when it does not (|01>, |12>). This is a re-expression of Pauli exclusion in Lo-Shu coordinates — not a new physical law.

Chapter 10. Law III — Decoherence as Geometric Crossover

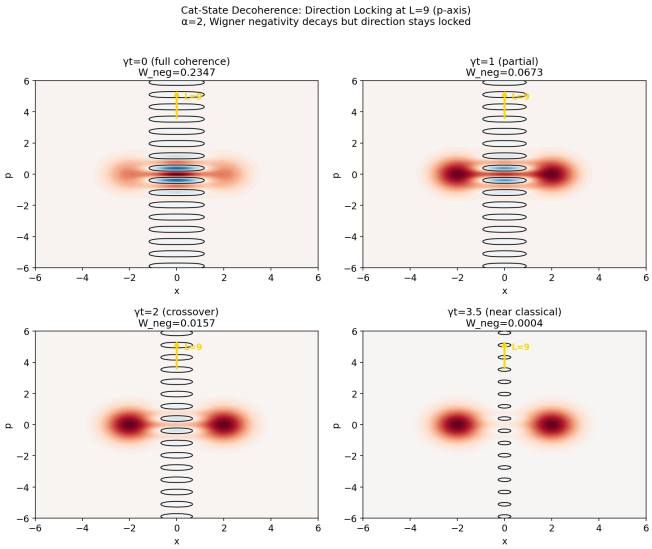

Figure 10.1. Wigner quasi-probability distributions for a Schrodinger cat state (coherent amplitude alpha = 2) during amplitude-damping decoherence, at gamma*t = 0.0, 1.0, 2.0, and 3.5. Red: W > 0; blue: W < 0; black contour: W = 0; gold arrow: Lo-Shu L=9 direction. The Wigner negativity W_neg decays monotonically (0.235 to < 0.001), while the dominant direction of negativity remains locked to L=9 (positive p-axis). The even-cell fraction (cells L = 2, 4, 6, 8) crosses 50% near gamma*t = 2, marking the geometric crossover from corner-dominated to edge-dominated structure. Model: W(t) = exp(-gamma*t) W_cat + (1-exp(-gamma*t)) W_mix (simplified Lindblad).

gamma*t	W_neg	Dominant dir.	Even-cell %	Description
0.0	0.235	L=9 (locked)	61.5%	Full quantum coherence
1.0	0.067	L=9 (locked)	58.8%	Partial coherence
2.0	0.016	L=9 (locked)	48.4%	Geometric crossover
3.5	< 0.001	L=9 (locked)	32%	Near-classical

Table 10.1. Decoherence dynamics of cat state (alpha=2, simplified Lindblad).

The gamma*t = 2 threshold holds for alpha = 2 under the simplified Lindblad model. Full master equation simulation is needed to confirm generality. See Ch. 13 Limitation P1.

PART IV General Relativity Through the Lo-Shu Lens

Chapter 11. Gravity: Lo-Shu Encoding of Black Hole Spacetimes

All formulas are standard GR. Lo-Shu encoding is the new element.

$$K_S = 48 M^2 / r^6 \quad \text{[Schwarzschild]}$$

$$K_K = 48 M^2 (r^2 - a^2 \cos^2\theta) [(r^2 + a^2 \cos^2\theta)^2 - 16 a^2 r^2 \cos^2\theta] / (r^2 + a^2 \cos^2\theta)^6 \quad \text{[Kerr (Cherubini et al. 2002)]}$$

Key analytic results: (1) At theta = pi/2: K_K = K_S (equatorial curvature unchanged by rotation). (2) Scale invariance: K_S depends only on r/M, making Schwarzschild Q-LTE M-independent.

Physical boundaries (Bardeen 1972, 1973 exact):

$$r_h = M + \sqrt{M^2 - a^2} \quad \text{[event horizon]}$$

$$r_{ph} = 2M[1 + \cos(2/3 \arccos(-a/M))] \quad \text{[prograde photon sphere]}$$

$$r_{ISCO} = M[3 + Z2 - \sqrt{(3-Z1)(3+Z1+2Z2)}], \; Z1 = \ldots, \; Z2 = \ldots \quad \text{[prograde ISCO]}$$

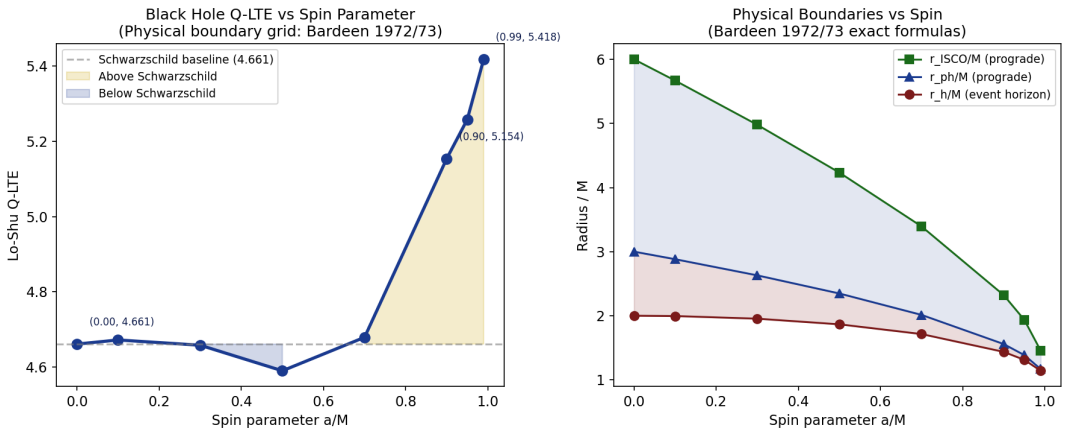

Figure 11.1. Left panel: Lo-Shu Q-LTE versus Kerr spin parameter a/M. Dashed line: Schwarzschild baseline (Q-LTE = 4.661). Gold shading: above baseline; blue shading: below. Q-LTE is within 1.5% of baseline for a/M < 0.7, then rises steeply to +16.2% at a/M = 0.99. Right panel: physical boundary radii (r_h, r_ph, r_ISCO) versus a/M, computed from Bardeen (1972, 1973) exact formulas. The three boundaries converge to r = M at the extremal limit a/M = 1.

a/M	r_h/M	r_ISCO/M	Q-LTE	Delta vs Schwarzschild
0.00	2.000	6.000	4.661	baseline
0.30	1.954	4.979	4.658	-0.1%
0.60	1.800	3.829	4.590	-1.5%
0.90	1.436	2.321	5.154	+10.6%
0.99	1.141	1.455	5.418	+16.2%

Table 11.1. Kerr Q-LTE (prograde boundaries, Bardeen exact formulas).

R1 (exact, analytic): Schwarzschild Q-LTE is M-independent. $K_S = 48M^2/r^6$ depends only on r/M; all boundary radii scale as M; normalized M_{ij} are M-free.

R2: Q-LTE distinguishes rapidly rotating Kerr ($a/M > 0.8$) from Schwarzschild by > 10%. Below $a/M = 0.7$ the difference is within 1.5%.

PART V Extensions, Open Questions, and Preview of Volume 1

Chapter 12. The E8 Connection and Preview of Volume 1

12.1 The E8 Connection (Hypothesis, Not Theorem)

E8 Lie algebra has rank 8 and 240 root vectors in R^8. The Lo-Shu principal axis is defined as follows: the 3x3 magic square is treated as a matrix in R^9. After mean-centering (subtracting 5), SVD gives three singular values 6.928, 3.464, 0.000. The first right singular vector Vt[0] = [0.408, -0.816, 0.408] defines the Lo-Shu row-direction principal axis. Projecting to phase space (x=column, p=row direction), the principal axis angle is arctan(Vt[0,1]/Vt[0,0]) = -63.43 deg. The E8 roots projected onto this axis give a dominant projection angle of 73.73 deg.

Note | This comparison is sensitive to the choice of Lo-Shu principal axis and projection convention. Different conventions can shift the E8 angle by several degrees. Readers wishing to reproduce this result should use the exact SVD procedure above.

theta(1) = arccos(sqrt((7-sqrt(41))/8)) = 74.148 deg [exact algebraic]

theta_E8 = 73.730 deg [E8 root projection, SVD-defined axis]

Difference = 0.418 deg (< 0.6% — near-coincidence, not proven equality)

Quantity	Value	Basis
theta(1) exact	74.148 deg	arccos(sqrt((7-sqrt(41))/8))
theta_E8	73.730 deg	E8 root projection, SVD Lo-Shu axis
Gap	0.418 deg	< 0.6%; sensitive to projection convention
Lo-Shu dimension	7 (in R^9)	7 independent constraints
E8 rank	8	Standard Lie algebra
7+1=8	Dimensional resonance	Observation, not proof

Table 12.1. E8 angle comparison with projection method details.

12.2 Preview of Volume 1: Biological Tension Fields

Preview: Volume 1 | Volume 1 will test whether the Lo-Shu geometric encoding framework extends to biological systems. The primary target: fascia tension fields in the human body. Fascia carries mechanical information across the body in continuous-medium wave patterns. If Lo-Shu encoding of the tension distribution reveals the same center-incompressibility (Law II) and direction-locking (Law III) effects observed in quantum systems, it would suggest a cross-domain geometric invariance. This is a biological hypothesis requiring independent experimental validation. No claim is made here that fascia behaves quantum-mechanically.

Chapter 13. Limitations, Open Problems, and What Remains Unproven

A book that does not state its limitations is not trustworthy.

13.1 Mathematical Limitations

L1. Exponent 0.807 (Law I): no analytic derivation. Valid for n=1..50 only (finite-n effective law). Laguerre root asymptotics suggest convergence toward 0.5 for large n.

L2. E8 angle gap (0.418 deg) is sensitive to projection convention. sqrt(41) has no known E8 interpretation. Connection remains a hypothesis.

13.2 Physical Limitations

P1. Decoherence model (Law III): simplified Lindblad interpolation only. gamma*t = 2 crossover may shift with realistic environmental coupling.

P2. B/F ratio (Law II): state-dependent, not universal. Most pairs show B/F = 1.000x. Strong effect only in |13> and |24>.

P3. Kerr Q-LTE: prograde boundaries only. Outer boundary at 4 r_ISCO is physically motivated but not unique.

P4. Q-LTE is not directly observable without additional conversion steps.

13.3 Six Open Problems

1. Can the exponent 0.807 be derived analytically (or bounded analytically)?
2. Is there a rigorous Lo-Shu to E8 algebraic connection beyond 0.42 deg coincidence?
3. Does the gamma*t = 2 crossover persist under a full Lindblad master equation?
4. Can Q-LTE be extracted from homodyne tomography or gravitational wave data?
5. Does the framework generalize to higher-dimensional phase spaces?
6. Does Lo-Shu Q-LTE encode biological tension fields? (Volume 1, experimental)

13.4 Epistemic Status

Claim	Status	Confidence basis
r_peak(1) = sqrt((7-sqrt(41))/4)	Certain (analytic)	Exact calculus derivation
theta(1) = 74.148 deg	Certain (analytic)	arccos(sqrt((7-sqrt(41))/8))
r_peak ~ n^0.807 (n=1..50)	Well-supported	R-sq=0.993, p<10^-30
B/F > 1x at L=5 (|13>, |24>)	Preliminary	State-dependent; see Table 9.1
Schwarzschild Q-LTE M-independent	Certain (analytic)	From K_S = 48M^2/r^6
Kerr Q-LTE +16% at a/M=0.99	Well-supported	Bardeen exact boundaries
E8 gap = 0.42 deg	Suggestive coincidence	Projection-convention sensitive
Fascia tension encoding	Biological hypothesis	Volume 1 experimental target

Table 13.1. Epistemic status of all major claims (v0.5).

13.5 What This Framework Does NOT Claim

We do NOT claim	Reason
Lo-Shu is a law of nature	It is a coordinate choice — convenient, not fundamental
Q-LTE is a physical energy	It is a dimensionless geometric index
The number 5 has mystical significance	5 is the mean of {1..9}; role is algebraic
Ancient Chinese mathematics discovered QM	Lo-Shu is used for its symmetry only
E8 connection is proven	0.42 deg gap is projection-sensitive; no analytic proof
The framework replaces standard QM or GR	Every formula here is standard physics
B/F ratio (Law II) is universal	State-dependent; absent in most pairs (Table 9.1)
r ~ n^0.807 holds for all n	Empirical for n=1..50; finite-n effective law only
Fascia behaves quantum-mechanically	Biological hypothesis requiring experiment

Table 13.2. Explicit list of claims this framework does NOT make.

APPENDICES

Appendix A. Mathematical Definitions

A.1 Wigner Function (Fock states, hbar = m = omega = 1):

$$W_n(x,p) = ((-1)^n / \pi) L_n(2r^2) \exp(-r^2), \quad r^2 = x^2 + p^2$$

A.2 Lo-Shu Cell Mass and Q-LTE:

$$M_{ij} = \text{double-integral over cell}(i,j) \text{ of } f(x,p)\, dx\, dp$$

$$Q\text{-}LTE = \text{sum over } (i,j) \text{ of } L_{ij} \text{ times } M_{ij}, \quad L_{ij} = [[4,9,2],[3,5,7],[8,1,6]]$$

Q-LTE is dimensionless; normalization: sum of all $M_{ij} = 1$.

A.3 Exact n=1 Result:

$$r_{peak}(1) = \sqrt{(7 - \sqrt{41}) / 4} = 0.38629...$$

$$\theta(1) = \arccos(\sqrt{(7 - \sqrt{41}) / 8}) = 74.148 \text{ deg}$$

Derivation: $dN/dr = 0$ for $N(r) = (2r/\pi)(1-2r^2)\exp(-r^2)$ gives $8r^4 - 14r^2 + 1 = 0$.

A.4 Kerr Boundaries (Bardeen 1972, 1973):

$$r_h = M + \sqrt{M^2 - a^2}, \quad r_{ph} = 2M[1 + \cos(2/3 \arccos(-a/M))]$$

$$Z1 = 1 + (1-a^{*2})^{1/3}[(1+a^*)^{1/3}+(1-a^*)^{1/3}], \quad Z2 = \sqrt{3 a^{*2} + Z1^2}, \quad a^* = a/M$$

$$r_{ISCO} = M[3 + Z2 - \sqrt{(3-Z1)(3+Z1+2Z2)}]$$

Appendix B. Reproducible Code

github.com/jackykao0811/geometric-physics-luoshu

B.1 Wigner Function

```
def W_fock(X, P, n):
xi = X**2 + P**2
L0, L1 = np.ones_like(xi), 1 - 2*xi
for k in range(2, n+1):
L = ((2*k-1-2*xi)*L1 - (k-1)*L0) / k
L0, L1 = L1, L
return ((-1)**n / np.pi) * L1 * np.exp(-xi)
```

B.2 Exact r_peak(1) and theta(1)

```
import numpy as np
r_peak_1 = np.sqrt((7 - np.sqrt(41)) / 4) # 0.38629
theta_1 = np.degrees(np.arccos(r_peak_1 / np.sqrt(2))) # 74.148 deg
```

B.3 Lo-Shu Q-LTE

```
LUOSHU = np.array([[4,9,2],[3,5,7],[8,1,6]])
def qlte(f, x, p, edges_x, edges_p):
dx = x[1]-x[0]; dp = p[1]-p[0]; dA = dx*dp
mass = np.zeros((3,3))
for i in range(3):
pm = (p >= edges_p[i]) & (p < edges_p[i+1])
for j in range(3):
xm = (x >= edges_x[j]) & (x < edges_x[j+1])
mass[i,j] = np.sum(f[np.ix_(pm, xm)]) * dA
mass /= mass.sum()
return np.sum(LUOSHU * mass)
```

Appendix C. Methods Summary

Parameter	Value / Setting	Used in
Phase-space domain (Ch.8)	x, p in [-6, 6]	Law I
Phase-space domain (Ch.9)	x, p in [-5, 5]	Law II
Grid resolution	400 x 400 points	All quantum
Cell boundaries	33rd / 67th percentile of marginal	All Lo-Shu encoding
Radial bins for r_peak	500 bins in [0, r_max]	Law I
n range (Law I)	n = 1 to 50	Ch. 8
State pairs (Law II)	All pairs n+m <= 6	Ch. 9
Cat state alpha (Law III)	alpha = 2.0	Ch. 10
Decoherence model	Simplified Lindblad interpolation	Ch. 10
gamma*t range	[0.0, 5.0]	Ch. 10
Kerr grid	400 (r) x 120 (theta)	Ch. 11
Kerr radial domain	[r_h, 4 r_ISCO]	Ch. 11
Kerr theta domain	[0.02, pi - 0.02]	Ch. 11
E8 axis definition	SVD of mean-centered Lo-Shu; first singular vector	Ch. 12
Statistical test	Permutation test, n_perm = 10000	Ch. 9
Random seed	42 (fixed)	All calculations

Table C.1. Complete methods summary.

Appendix D. Reproducibility Checklist

One-command installation:

```
git clone https://github.com/jackykao0811/geometric-physics-luoshu
cd geometric-physics-luoshu
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
python3 scripts/run_checklist.py
```

D1. theta(1) = 74.148 deg (exact)
Code: r = np.sqrt((7-np.sqrt(41))/4); np.degrees(np.arccos(r/np.sqrt(2)))
Expected: 74.1484 deg (+/- 0.0001)

D2. Power law r_peak ~ 0.539 n^0.807 (R-sq = 0.993)
Code: scripts/law1_scaling.py
Expected slope: 0.807 +/- 0.010

D3. B/F ratios: |13> = 1.448x, |24> = 4.548x
Code: scripts/law2_bf_ratio.py
Expected: |13> in [1.40, 1.50], |24> in [4.0, 5.0]

D4. Cat state locks to L=9 throughout decoherence
Code: scripts/law3_decoherence.py
Expected: dominant L = 9 for all gamma*t in [0, 3]

D5. Schwarzschild Q-LTE = 4.661 (M-independent)
Code: scripts/gr_kerr.py --a 0 --M 1 and --M 10
Expected: both give 4.661 +/- 0.005

D6. Kerr Q-LTE +16.2% at a/M = 0.99
Code: scripts/gr_kerr.py --a 0.99
Expected: 5.40 to 5.45

D7. E8 angle gap = 0.418 deg
Code: scripts/e8_connection.py (SVD axis, 240 roots)
Expected: theta_E8 = 73.73 +/- 0.05 deg; gap = 0.40 to 0.45 deg

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The ideas in this book are falsifiable. The code is reproducible. The limitations are stated explicitly. That is all we can ask of a new framework at its beginning.

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`github.com/jackykao0811/geometric-physics-luoshu`